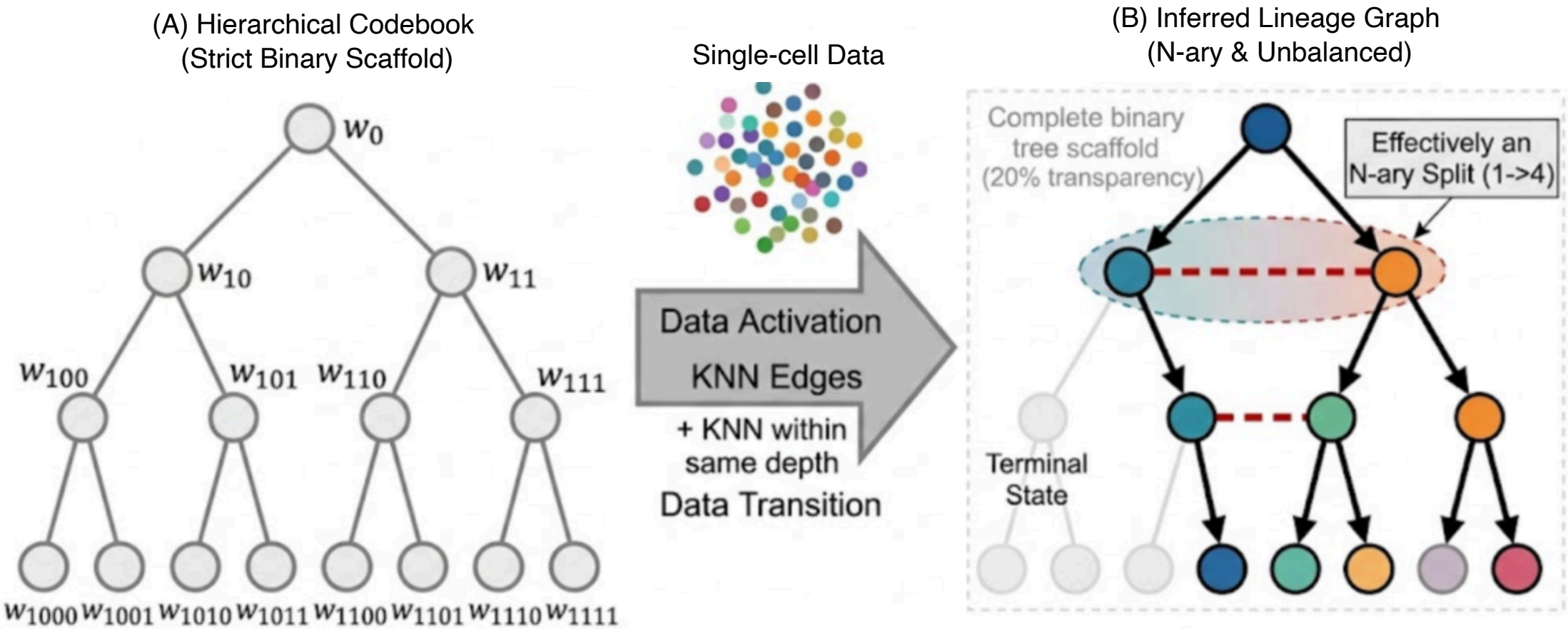

Figure R2: *Illustration of HDTree's transition from a rigid binary scaffold to a flexible biological graph.* ***(a)*** *The underlying Hierarchical Tree Codebook remains a strict binary structure for computational efficiency.* ***(b)*** *By activating only specific paths (handling unbalanced depths) and linking sibling nodes via horizontal KNN edges (red dashed lines), highly similar binary nodes form biological "macro-states" (shaded region). This mathematically preserves the binary codebook while effectively modeling complex* ***multi-branching (n-ary) biological lineages*** *(e.g., effectively a 1-to-4 split).*